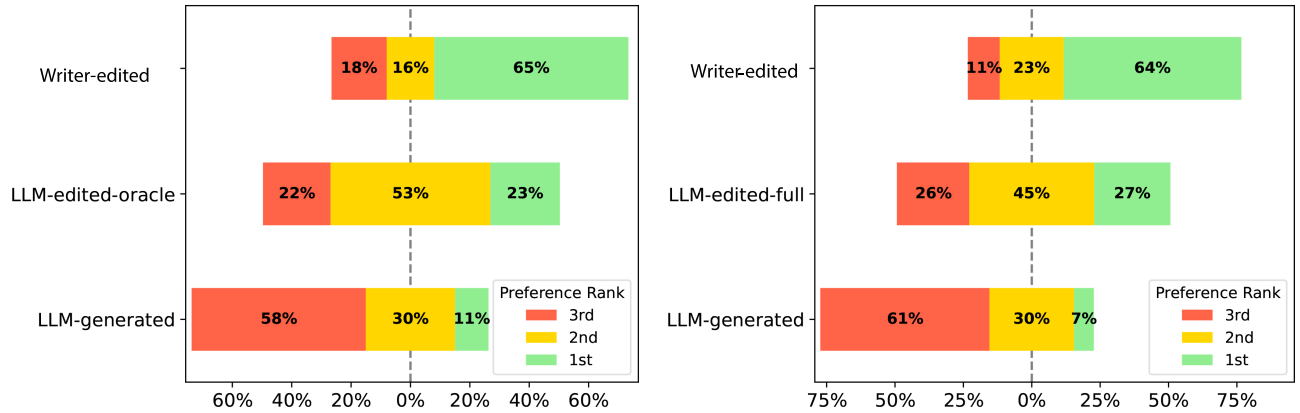

(a) Samples involving oracle detection (N=300) **(b) Samples involving automated detection (N=300)**

Figure 8: Distribution of rankings for each variant in the preference annotation study. Annotators read three variants of a paragraph (Writer-edited, LLM-generated, and either LLM-edited-oracle or LLM-edited-full) and ranked them by preference (1st, 2nd, 3rd). The distribution indicates how often each variant was ranked as best (1st), second best (2nd), or worst (3rd).

preference rankings. To ensure the validity of the results, **no participant reviewed paragraphs they had seen or edited in past tasks, and only judged paragraphs edited by other experts**.

To analyze the reliability of the results we calculate inter-annotator agreement using Kendall's W (also known as Kendall's coefficient of concordance) [35] which ranges from 0 (no agreement) to 1 (complete agreement) to evaluate agreement amongst participants. Our annotation achieves an overall agreement of **0.505**, suggesting a moderate level of agreement across all participants. This moderate agreement underscores the subjective nature of judging writing quality while suggesting that certain differences are distinctive enough to be consistently preferred by multiple participants.

Table 12 and Figure 8 summarize the preference evaluation results, showing average ranks across 600 annotations. Overall, the **Writer-edited condition is most preferred, a sign that expert-edited text is unrivaled in terms of writing quality**, being marked as the most-preferred paragraph variant 65% of the time and achieving an average rank of 1.5. Next, the LLM-edited variants come in second, with an average rank of 1.99 for both the LLM-edited-oracle and LLM-edited-full conditions. Surprisingly, the condition that leveraged the oracle span from writers ranks almost identical to the condition with automatically detected spans. This provides evidence that detection of problematic spans is not the bottleneck in improving writing quality, and instead the **rewriting module (which is common to both conditions) is what dictates the overall performance of an automated text-editing pipeline**. Finally, the original **LLM-generated** paragraphs achieve the worst ranking performance, being least preferred 60% of the time, and achieving an average rank of 2.51.

In summary, our experiment validates the potential benefit of automatic editing to improve writing quality: although automatic editing does match the quality of edits provided by professional writers, LLM-edited text is significantly preferred to LLM-generated text by expert writers (Design Principle 3). In other words, this experiment shows that **LLMs can improve the quality of their writing in a fully automatic way, by first generating a draft, selecting problematic spans, and then rewriting such spans**. These results align with previous findings [32, 40, 89, 114] showing that iterative refinement and editing can improve LLM outputs.

6.4 Qualitative Insights into LLM edits

Given how LLM-edited paragraphs are often preferred over default LLM-generated ones and sometimes even Writer edited paragraphs, we inspect if there are specific categories of edits where LLMs perform better, more so if that caters to a certain genre. Table 13 shows examples of original spans edited by LLMs that vary across genre and category. In general, there isn't a domain or category where LLM edits are better than the rest except for `Purple Prose`. To edit `Purple Prose` models mostly need to learn how to simplify the text (i.e., generate a paraphrase in simple language). LLMs are effective at Sentence Simplification [28] so this doesn't come across as a surprising finding. Categories like `Poor Sentence Structure` and `Unnecessary Exposition` deliver mixed results. Sometimes they are effective. For instance Row 7 in Table 13 shows how the model improves the structure by breaking the run-on very long sentence. However, for the other example in Row 6, the human edit is *Then he hands his card, always leaving a 22% tip* while the model edit is a mere paraphrase. Similarly, under the `Unnecessary Exposition` category, the LLM is very good at editing sentences with the following structure [Main clause] [comma] [exposition] where they simply remove the exposition (as learned from Writer edits). However, the bigger issue in this category is revealing the subtext. When a given input span doesn't conform to the structure above or consists of multiple sentences (Row 9, Table 13) LLMs don't align with human edits. For `Cliché`, models mostly rewrite it in simpler language typically with fewer words which while being effective isn't what humans do. This also shows the fundamental overlap between `Cliché` and `Purple Prose` as shown in Section 5.2. Finally, models are mostly ineffective at addressing `Lack of Specificity and Detail` (more details in section 7.2). These findings demonstrate the challenge of editing text across any domains

Genre	Category	Original Span	LLM edited span
Literary Fiction	Lack of Specificity and Detail	seems to embody the city's frustrations and disappointments	seems to embody the city's frustrations and disappointments, his eyes a deep well of anger and desperation, his face a topographic map of the city's corruption and neglect.
Food Writing		Gone are the individual ramekins, replaced by a single, generous vessel that invites sharing.	Gone are the individual ramekins, replaced by a deep-dish pie plate that invites communal indulgence. The golden crust cradles a sea of silky custard, its surface a glassy expanse of caramelized sugar.
Literary Fiction	Purple Prose	confusion clouded his understanding like the haze hanging over the park	he remained confused
Food Writing		It's the democratization of ingredients that sets the bánh mì apart	It's the variety of ingredients that sets the bánh mì apart
Internet Advice		But pain has a way of cracking us open, making space for new possibilities we couldn't have imagined before.	But pain can open us up to new possibilities.
Travel Writing	Poor Sentence Structure	He tips generously but modestly, as if acknowledging a pact of respect	He tips generously yet with restraint, honoring a mutual respect
Literary Fiction		The gulf between her vision for her future and her parents' hopes wasn't something that could be bridged with words alone, and Bella realized that sometimes growth requires difficult choices.	Bella realized that her vision for the future and her parents' hopes were irreconcilable. Growth, she understood, often demands difficult choices.
Internet Advice	Unnecessary Exposition	, a silent reminder of everything it once meant	-
Personal Essay		Now, when life deals its inevitable blows, I think of those stones. I remember that sometimes it's the cracks that let the light in, and that even the most jagged edges can be smoothed by time and patience.	Now, when life deals its inevitable blows, I think of those stones.
Travel Writing	Cliché	As the weekend comes to a close, reflect on the kaleidoscope of experiences you've had, and how each neighborhood has contributed to the vibrant tapestry that is Los Angeles.	As the weekend comes to a close, you'll have a new appreciation for the city's diversity.
Food Writing		a dish that feels both familiar and unexpected, much like discovering a hidden alley in a well-known city. It's	a familiar dish with a surprising twist. It's
Literary Fiction		, watching his chest rise and fall with labored breaths	-
		I noticed a vase of wilting flowers on the nightstand, forgotten in the weight of more pressing concerns	The flowers on the nightstand had wilted

Table 13: LLM edited spans from different categories across different sub-genres of writing

that require deeper emotional resonance or cultural commentary and the importance of rethinking the design of alignment for tasks with subjective or fuzzy rewards.

7 DISCUSSION

7.1 How is editing human writing different from LLM-generated text?

Editing human writing and LLM-generated text presents distinct challenges and requires different approaches. Human writing often contains nuanced expressions, personal style, and contextual references that reflect the author's unique voice and experiences. Editors must preserve these elements while refining clarity, structure, and coherence. In contrast, LLM-generated text may lack consistent tone and exhibit repetitive patterns (Section 5). We asked writers to explain the differences in editing LLM-generated text compared to Human-written text. Several writers mentioned that LLM writing often required more extensive editing, mainly to remove unusual and sometimes nonsensical metaphors, inappropriate use of complex vocabulary that doesn't fit the context, and improving an

overall tone that comes across as impersonal and mechanical. In an exchange, W3 noted "*I edit a lot of prose for my magazine but one thing that stuck with me as I was editing these paragraphs are the massive amount of cliché, histrionic descriptions, and direct exposition of intended meanings rather than effective representation. Indeed very strange*". They observed that the types of edits needed for LLM-generated text were often similar, but the sheer volume of necessary changes was higher than that of human writing. LLM-generated content's repetitive nature paradoxically made the human editors feel robotic while trying to improve it. By the edits that catered to deep emotional resonance or cultural commentary,

7.2 How well can LLMs mimic edits from writers?

Our preference ranking results in Section 6.3 indicate that automatically edited paragraphs frequently rank second and sometimes first. This raises questions about LLMs' ability to analyze textual patterns and generate content closely resembling a given writer's edit. For the span "Janet lay in bed each night, her mind a whirlpool of restless thoughts", both LLM and writer identified it as cliché. The LLM edited it to "Janet lay in bed each night, unable to sleep" while the writer changed it to "Each night, Janet lay prone in her bed and unable to sleep". LLMs can also split run-on sentences and improve poor structure. For "Sarah froze, realizing it was her high school sweetheart, Alex, whom she hadn't seen in over a decade", the LLM edited it to "Sarah froze. It was Alex, her high school sweetheart. She hadn't seen him in over a decade", similar to the writer's edit. However, LLMs sometimes replace clichés with other clichés or fail to remove unnecessary exposition. The most challenging edit category is Lack of Specificity and Detail where LLMs often fail to add engaging details. For "Her irritation slowly morphed into a strange, disconnected calm", the writer added "After all, the noise just meant that she wasn't the only one awake at this hour." The model's edit was less effective: "Her irritation slowly morphed into a strange, disconnected calm. The repetitive thump-thump-thump became almost hypnotic, lulling her into a trance-like state". Much evocative detail in human writing comes from lived sensory experiences. LLMs on the other hand lack the grounded understanding that helps humans select vivid, emotionally resonant details. Additionally, when we combine autoregressive objectives with post-hoc adjustment through RLHF (typically designed to prevent toxic/harmful text generations) models often default to common, generic descriptions rather than specific ones since those are "safer" predictions. However, we note one potential limitation in our experiments is our reliance on few-shot instructions, requiring the model to learn rewriting from only a few examples. Training on the entire LAMP Corpus or more data might improve edit quality.

7.3 What recommendations can we provide for future LLM-based writing support tools that aspire to improve the co-writing experience?

Eminent author Curtis Sittenfeld calls LLM writing the literary equivalent of fat-free cookies [12]. LLMs are proficient at producing sentences that are grammatically correct and devoid of spelling errors. Beyond that, LLMs require extensive learning to effectively assist humans in improving their writing. In his essay *Politics and English Language* [85], George Orwell said "Never use a long word when a short one will do." LLM writing transgresses this simple rule by overusing lofty words. Clichés are bound to slip into even the best human writing, but when it comes to LLMs it simply cannot write without them. We believe this is partially a drawback of the technology behind LLMs. When an LLM calculates the probability of one word following another, clichés become very likely, because they've appeared so many times before. This explains why every other generated response is rife with clichés despite our prompt explicitly asking LLMs to avoid clichés and overused tropes (Table 14). LLMs need to learn how to identify and write without clichés such that it is engaging to every single reader. Overwriting is a bigger problem than underwriting. The rule for most writers is, "If in doubt, cut it." [46] The Pulitzer Prize-winning writer John McPhee has called the process "writing by omission." [75]. To become a better writer LLMs need to learn how to avoid unnecessary exposition. Last but not least, structure is what good writing hangs on [46]. Long, run-on sentences are hard to read, and LLMs need to know when and how to split effectively to better manage flow and clarity.

7.4 What are the potential long-term effects on language evolution and writing styles as LLM becomes more prevalent and how can aligned editing tools help?

The increasing prevalence of large language models (LLMs) could significantly impact language evolution and writing styles over time. There's potential for more homogenized writing as people rely on LLM-generated content, possibly leading to a reduction in linguistic diversity and individual voice [105]. However, well-designed editing tools aligned with expert writing practices could help counteract these effects. Such tools could encourage more nuanced and sophisticated language use, preserve stylistic diversity, and promote critical thinking about word choice and sentence structure. By highlighting elements of expert writing, these tools could elevate overall writing quality while still allowing for personal expression, potentially steering language evolution towards greater clarity, precision, and effectiveness in communication [68].

7.5 Why are there no significant differences in perceived writing quality or types of edits needed across texts generated by different large language models?

The lack of significant quality differences between LLM writing is somewhat unexpected and warrants deeper examination. Several factors however may explain this surprising finding. Books play a crucial role in the training of generative AI systems. Their long, thematically consistent paragraphs provide information about how to generate coherent and fluent text. All three models (GPT-4, Claude 3.5, Llama 3.1) are pre-trained on Books3corpus [54] which constitute a bigger portion of Pile [41] pre-training data. Additionally, all LLMs rely on Scale AI for collecting preference data and there is

[12] https://www.nytimes.com/2024/08/20/opinion/beach-read-ai.html